

Ground floor study

Where the garage stair lands, and the floor planned around it: a tall living room behind a pentagon of glass with a fireplace at its side, a quarter-turn stair folded into the back corner with the toilet in its crook, and the metre the house grew at the back given to the living room. Metres, to scale, house frame: x along the house (the valley at the bottom), z across (north to the right).

$$9.00 \times 7.00$$

over all, a metre further into the hill; 8.40×6.40 clear, 53.8 m^2

$$5.00 \times 5.20$$

the living room, the whole front from the bedroom wall to the glass: the extra metre is its

$$\times 5.70$$

where the garage flight lands: the back wall line, 1.30 of landing to the stair up

$$4 + 3 + 9$$

the stair up: a quarter turn in the north-west corner, 17×17.1 on 0.27 treads, winders at the corner, the toilet in its crook

$$6.40 \times 9.10$$

the panorama: one pentagon of glass, 5.90 at the walls, 9.10 at the apex, the living room's alone

$$9.90$$

the chimney, up through the north slope at the glass's right; the hearth 1.40×1.20 with the fire facing the room

$$26.0 \cdot 11.2 \cdot 3.2 \cdot 2.2$$

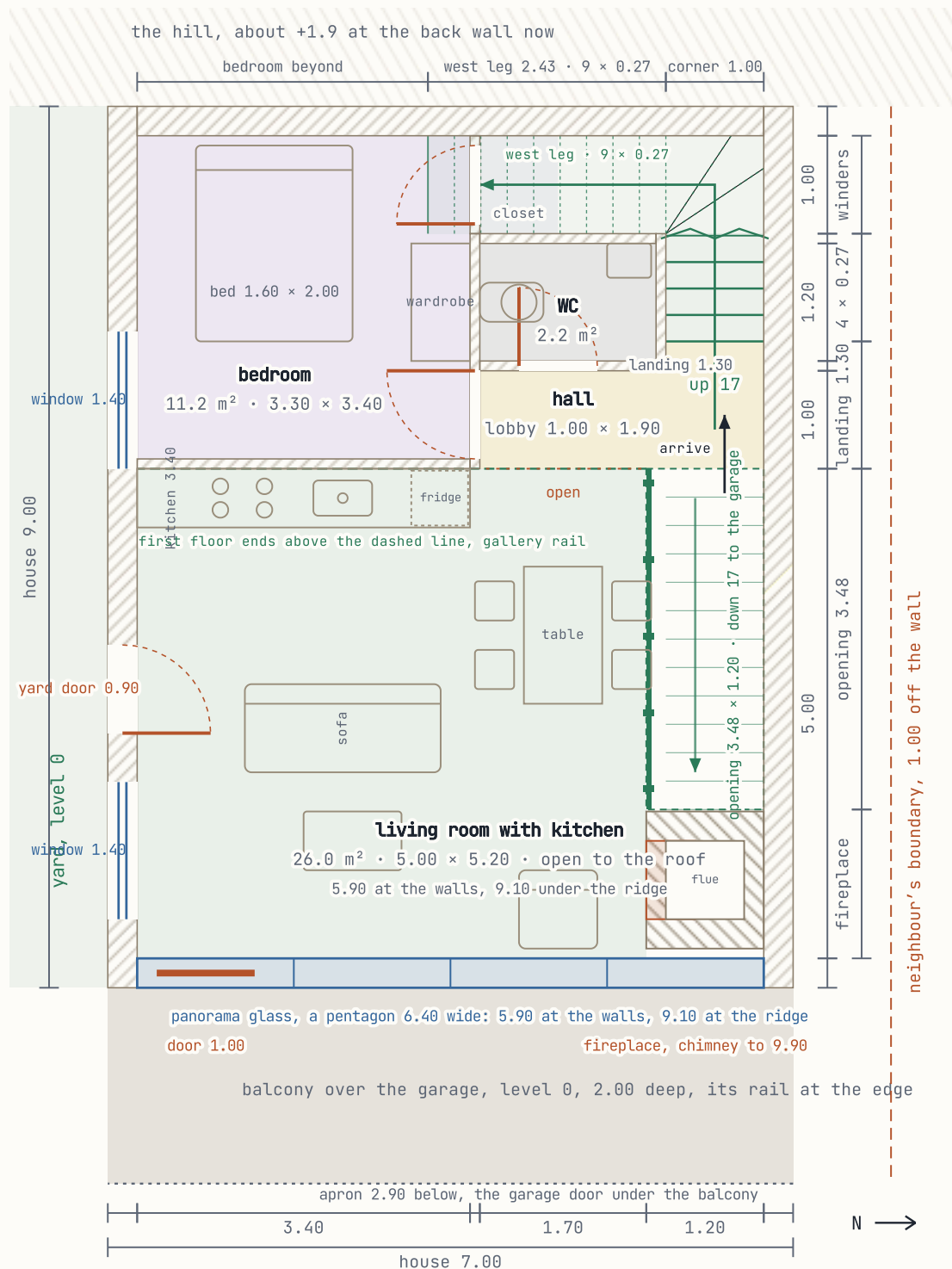
living room with kitchen, bedroom, hall, toilet, m^2

$$18.3$$

first floor behind the tall room, m^2 clear: 3.40×6.40 less the stairwell

Plan, 1 m above the floor

The hill is at the top, the valley and the balcony at the bottom, the yard to the left (south), the neighbour's boundary 1 m off the right (north) wall. Walls 0.30, partitions 0.10.



- hall
- toilet
- bedroom and its closet under the stair
- living room with kitchen, open to the roof
- glass
- fireplace

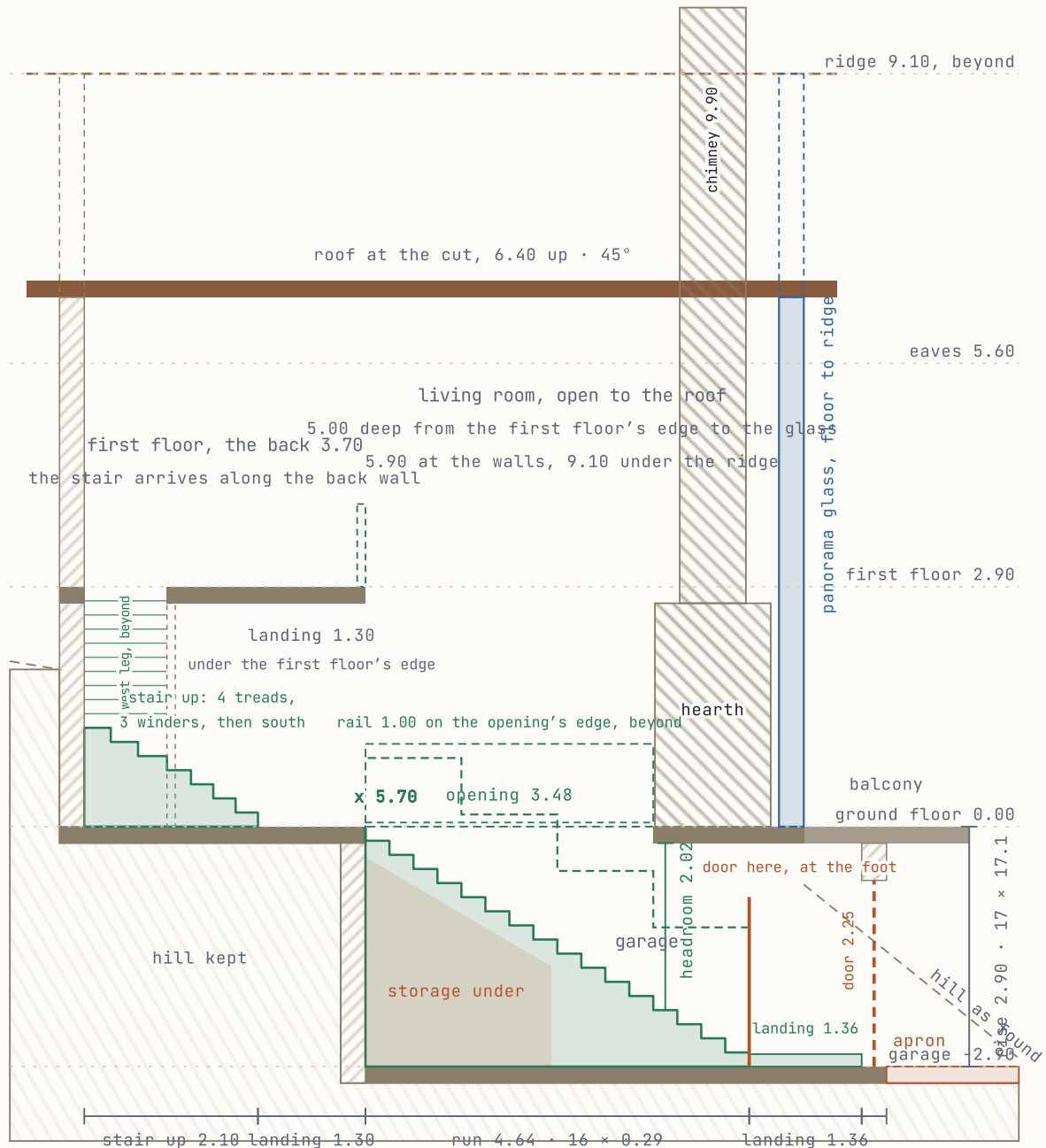
- **The living room has the metre.** It is the whole front from the bedroom wall to the glass, 5.00 × 5.20, and it is tall: the first floor stops at x 5.7, so from the kitchen to the glass the room rises past the first floor to the roof — 5.90 at the walls, 9.10 under the ridge. Its east wall is the panorama, and nothing

else stands behind it: one pentagon of glass, 6.40 wide, from the floor to the roof, with a 1.00 door to the balcony at its south end.

- **The fireplace** stands at the glass's north end, in the corner between the garage opening and the glass: a masonry hearth 1.40×1.20 with the fire facing south into the room, 0.80 wide and 0.35 off the floor, and a 0.80×0.80 chimney up through the north slope to 9.90, 0.80 over the ridge. Its west side closes the garage opening's east end.
- **The hall is where the stair lands.** You come up facing the hill at x 5.7, with 1.30 of landing, and the stair up carries straight on: four treads west along the north wall, three winders in the corner, nine treads south along the back wall. The lobby south of the landing, a metre wide and open to the living room, has the toilet door on its west side and the bedroom door at its end.
- **The stair up** is a quarter turn folded into the north-west corner, 17 risers of 17.1 on 0.27 treads, 1.00 wide, the toilet in its crook. Its footprint is 4.5 m^2 of treads, nothing more; the walls on its inner sides are the toilet's north and west walls, and under its west leg is a closet for the bedroom, 1.00×0.80 , 1.9 to 2.4 high. It arrives on the first floor at z 3.73 along the back wall, with a 1.00 landing beyond.
- **The entrance from the yard** is a 0.90 glazed door in the south wall beside the kitchen; a 1.40 window further along lights the sofa.
- **The kitchen** stands against the bedroom wall, 3.40 of counter with the fridge at the hall end. The table sits by the garage opening, the sofa faces the glass and the fire.
- **The bedroom** is 3.30×3.40 in the south-west corner with a 1.40 window to the yard: a 1.60×2.00 bed with its head to the back wall, a wardrobe on its north wall, the closet under the stair beside it; the stair's last treads pass over its north-west corner at 2.4 and more. Its back wall is against the hill.
- **The first floor keeps the back 3.70 m** (x 2–5.7): 3.40×6.40 clear less the 3.4 of stairwell, 18.3 m^2 — a landing, the bathroom over the toilet, a bedroom — with a gallery rail on its edge over the living room, and the attic over it.

Where the garage stair lands: section along it, looking south

Cut 0.8 m from the north wall, through the garage flight. Garage floor -2.90, ground floor 0.00, first floor 2.90, eaves 5.60, ridge 9.10. The roof is cut where it is 6.40 up; the ridge is beyond. The cut passes through the stair up's first leg and winders at the back, and through the hearth and chimney at the glass.

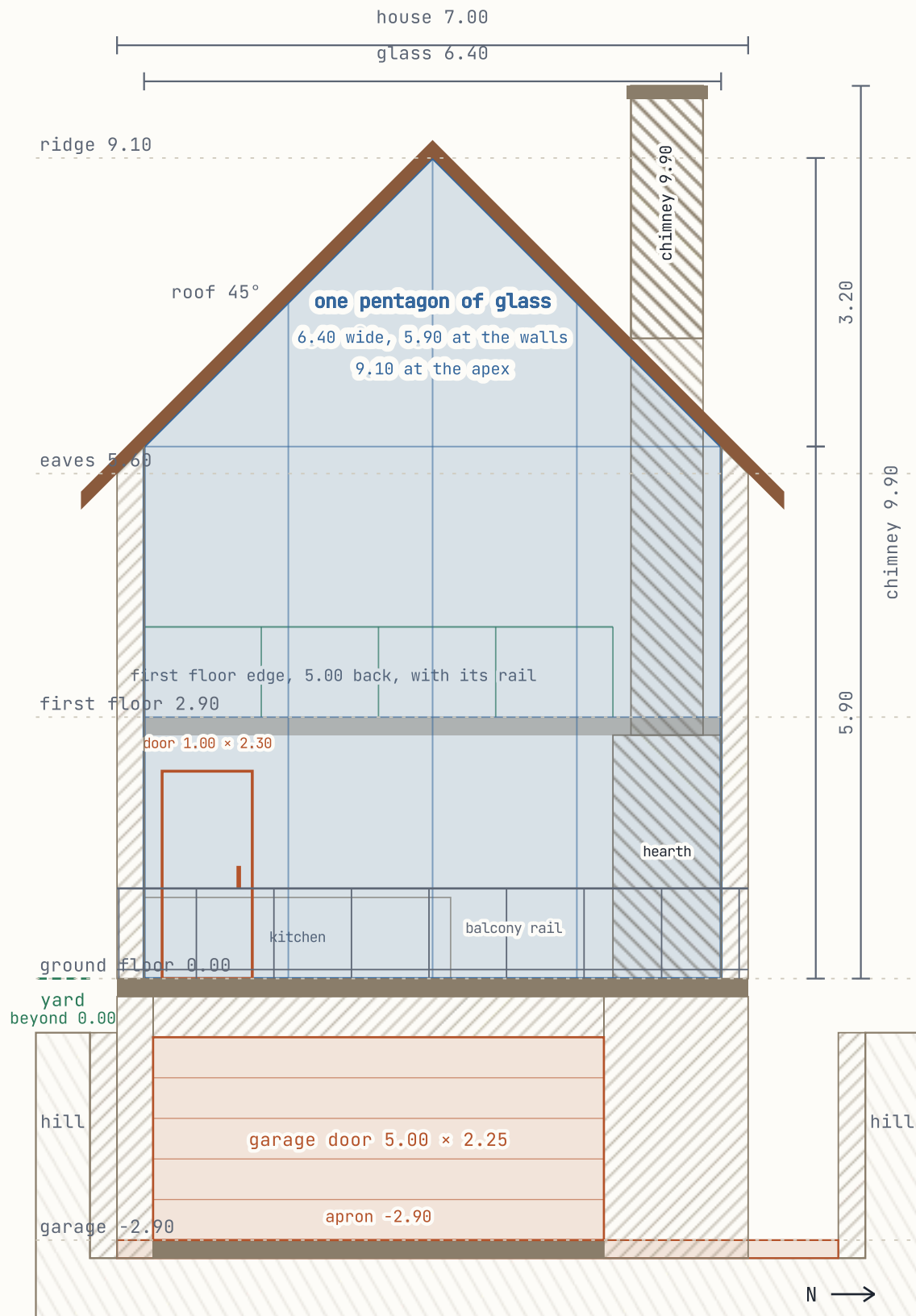


- **The top riser is the garage's back wall line, x 5.7:** the 17th step up lands on the ground floor over the garage's west wall, facing west with 1.30 m of landing before the stair up begins — and the first floor's edge is right above it, so you arrive at the foot of the tall room. The stair up carries the climb on along the north wall, round the corner and south along the back wall.
- **The opening** runs from the 5th riser to the top, 3.48 × 1.20 along the north wall, in the ground floor's deck; above it is the tall room. The deck stays over treads 1 to 4, and the hearth stands on it, closing the opening's east end.
- **Headroom is 2.00** over the 4th tread, the minimum. The stair up has the same: its first four treads sit under the first floor's deck, and from the fifth riser on, at the corner, the deck is open.

- **The door between garage and house** belongs at the foot of the garage flight, on the 1.36 landing inside the garage door, swinging onto that landing; a door at the top would open onto the stair.
- **The chimney** rises from the hearth through the roof's north slope, where the slope is 6.10 to 6.90 up, and on to 9.90: 0.80 above the ridge, as a flue 2.6 m off the ridge wants.

East face, from the valley: the pentagon

Looking west at the glass, north to the right as in the plan. The garage door under the balcony, the balcony at the living room's floor, the glass from there to the ridge, the chimney at its right.



- **One pentagon of glass**, 6.40 wide between the north and south walls, 5.90 at the walls, 9.10 at the apex: about 47 m² of glazing, split by mullions every 1.60 and transoms at the first floor's level and the

eaves line.

- **Seen through it:** the hearth and chimney at the north end, the first floor's edge 5.00 back with its rail, the kitchen.
- **The chimney** leaves the north slope near the eaves and stands 9.90 high, 0.80 over the ridge.
- **The balcony door** is at the south end of the glass, 1.00 × 2.30, where the balcony and the yard meet at the house corner.

Rooms

ROOM	CLEAR	M ²	LIGHT AND DOORS
Living room with kitchen	5.00 × 5.20	26.0	open to the roof, 5.90 to 9.10; the glass pentagon east with a 1.00 door to the balcony; the fireplace at the glass's north end; yard door 0.90 and window 1.40 south; open to the hall
Fireplace	1.40 × 1.20	1.7	the hearth, between the garage opening and the glass; chimney to 9.90
Bedroom	3.30 × 3.40	11.2	window 1.40 to the yard; door 0.90 from the lobby; the closet off its north wall
Closet under the stair	1.00 × 0.80	0.8	from the bedroom; 1.9 to 2.4 high
Hall: landing and lobby	1.30 × 1.00 + 1.00 × 1.90	3.2	the garage flight lands in it, the stair up starts in it; open to the living room
Toilet	1.20 × 1.80	2.2	door 0.80 from the lobby; no window, mechanical vent; in the crook of the stair
Stair up	4 treads + 3 winders + 9 treads	3.7	the closet is under its west leg; its last treads pass over the bedroom's corner
Garage opening	3.48 × 1.20	4.2	the garage flight below, railed on its south edge
Partitions		1.3	
Inside the outer walls	8.40 × 6.40	53.8	

Still open

- The first floor: 3.40 × 6.40 clear behind the tall room, less the stairwell, 18.3 m² — the stair arrives along the back wall at z 3.73, the bathroom goes over the toilet

beside it and the rest is a bedroom; the attic over it is another room if the roof is kept at 45° . A gallery out over the living room would win floor back at the cost of the tall room's depth.

- The fireplace: a wood stove in the same hearth would need a smaller flue; the chimney passes the roof near the eaves, 2.6 m off the ridge, so it stands 3.0 m above the slope and wants bracing.
- The back wall now stands a metre deeper in the hill (about +1.9 of ground against it), and the yard's west edge followed it to x 2: more retaining wall and more cut.
- The roof pitch: 45° drawn, which puts the apex at 9.10. At 35° it is 8.05 and the glass loses a metre.
- 47 m^2 of east glass: heat in winter and sun in summer want a plan — triple glazing, and shading or a deep roof overhang at the gable.
- The door at the foot of the garage flight: self-closing, fire-rated where the rules ask.
- A north window: the wall is 1 m from the boundary; the rules decide.
- Whether the outer walls sit inside the 9×7 (drawn) or outside it: outside would give 9.00×7.00 clear and push the east glass onto the balcony's first 0.30.